

Part A

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Signature and Date: Nishat Sharmin
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Ans to the Qus No 1 (a)

②

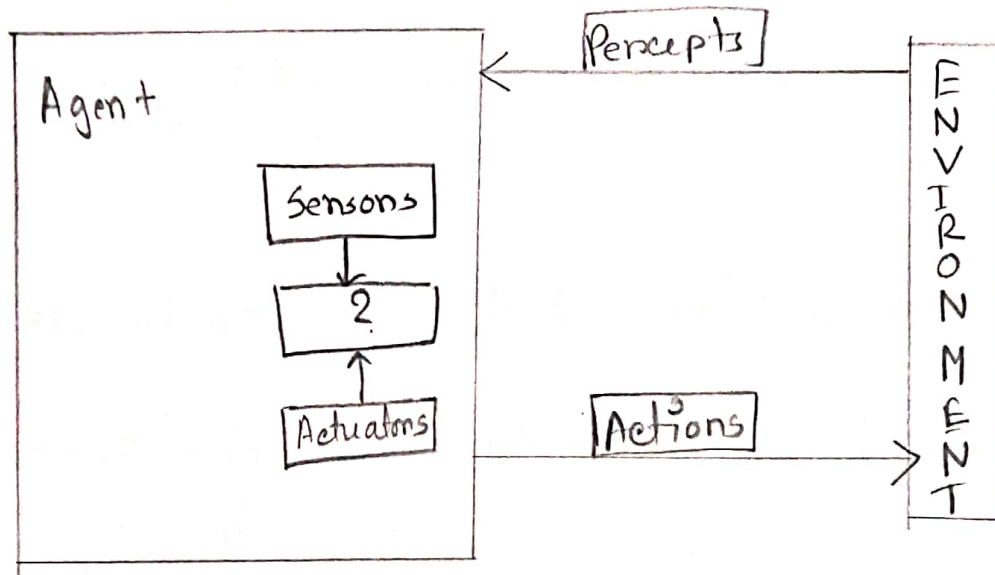


Fig: structure of Agent (Autonomous vehicle)

Structural components:

Sensors: for sensing obstacles and collecting data.

Actuators: To navigate and avoid obstacles.

Functional components:

Agent Program:

while switched on, perform sequentially the actions returned by the following function:

function Agent1 (Location, status)

if status = have-data then Action $\leftarrow$ PickData

else if status = have-rock then Action $\leftarrow$ Jump through
Rock

else if status = End then Action $\leftarrow$ Halt

else Action $\leftarrow$ Move to Right

return Action

Ans to the Ques No 1(b)

For applying the chaining algorithms on a knowledge base. The knowledge base should be represented logically i.e. PL and should contains only horn clauses.

Now given facts and rules are -

A

B

C

D

$$A \wedge B \Rightarrow R$$

$$A \wedge C \Rightarrow S$$

$$A \wedge P \Rightarrow E$$

$$B \wedge D \Rightarrow Q$$

$$D \wedge E \Rightarrow G$$

$$C \wedge Q \Rightarrow K$$

$$D \wedge F \Rightarrow M$$

$$D \wedge K \Rightarrow P$$

$$D \wedge P \Rightarrow F$$

$$P \wedge R \Rightarrow H$$

R (New)

Q (")

K (")

P (")

F (")

H (")

M (")

Now, $KB \models M$?

Execution of FE

A

B

$$A \wedge B \Rightarrow R$$

Modus ponens

$$\therefore KB \models R \text{ (added to KB)}$$

B

D

$$B \wedge D \Rightarrow Q$$

Modus Ponens

$$\therefore KB \models Q \text{ (added to KB)}$$

C

Q

$$C \wedge Q \Rightarrow K$$

Modus Ponens

$$\therefore KB \models K \text{ (added to KB)}$$

D

K

$$D \wedge K \Rightarrow P$$

Modus Ponens

$$\therefore KB \models P \text{ (added to KB)}$$

D

P

$$D \wedge P \Rightarrow F$$

Modus Ponens

$$\therefore KB \models F \text{ (added to KB)}$$

R

P

$$R \wedge P \Rightarrow H$$

Modus Ponens

$$\therefore KB \models H \quad (\text{added to } KB)$$

Q

Q

$$Q \wedge Q \Rightarrow$$

D

F

$$D \wedge F = M$$

Modus Ponens

$$\therefore KB \models M \quad (\text{added to } KB)$$

found

Ans to the Qus No 2

a) From the given table,
we can draw a weighted undirected graph
along with the heuristic values of nodes as
follows:

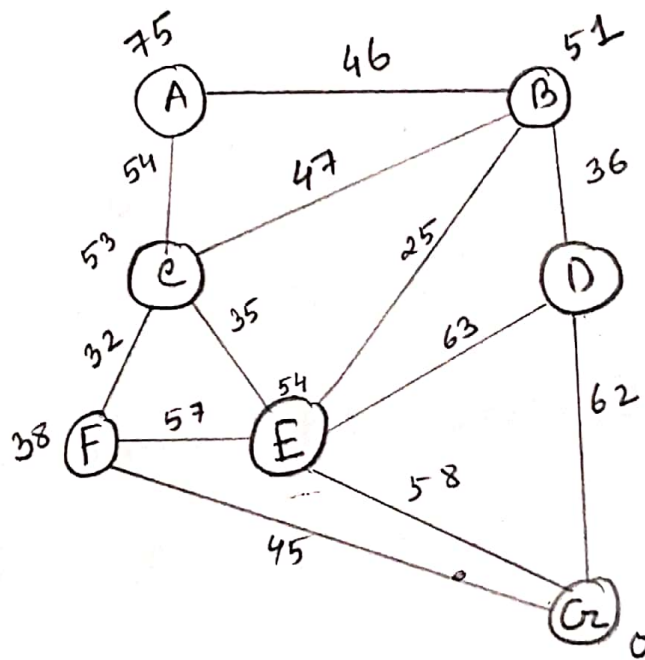


Fig. A weighed undirected graph
along with heuristic values

Given:

A is the initial node
and B is the goal node

Now,

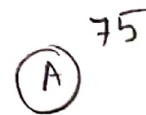
Applying Greedy Best-First Search algorithm

we get,

Step-1

PQ : [node (A, 75)]

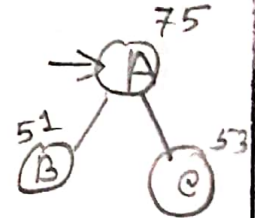
PP : []



Step-2:

PQ : [node (B, 51), node (C, 53)]

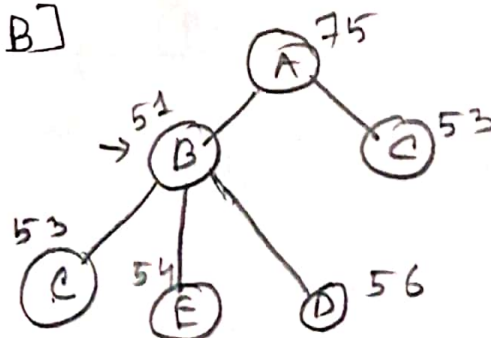
PP : [A]



Step-3:

PQ : [node (C, 53), node (E, 54), node (D, 56)]

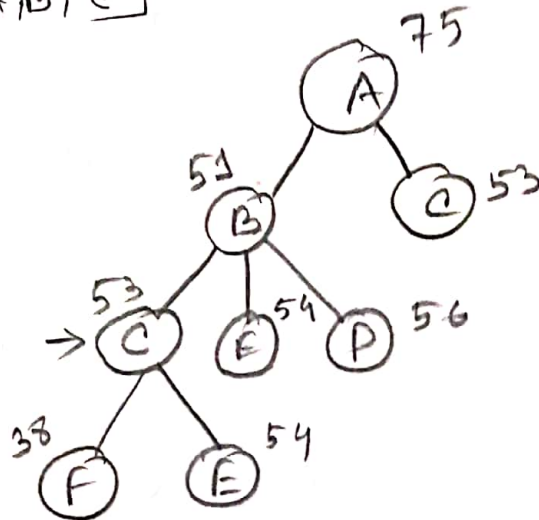
PP : [A, B]



step-4:

PQ : [node (F, 38), node (E, 54), node (D, 56)]

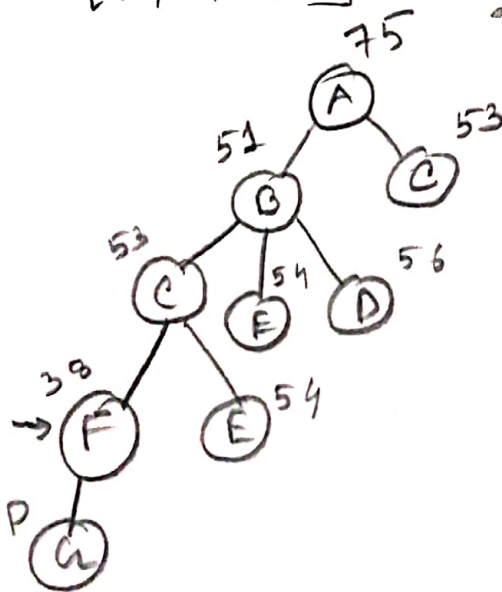
PP : [A, B, C]



step-5:

PQ : [node (G, 0), node (E, 54), node (D, 56)]

PP : [A, B, C, F]



Hence, the possible path is: $A \rightarrow B \rightarrow C \rightarrow F \rightarrow G$

$$\begin{aligned}\text{Total cost} &= 46 + 47 + 32 + 45 \\ &= 170\end{aligned}$$

Ans to the Ques No 2 (b)

There are many odds of the environment of an optimization problem containing various types of local maxima in it.

Presence of various local maxima creates a very big problem in any optimization technique specially when the optimization is set out to find the global optimum.

Besides after getting the local maxima, most of the case the optimization sticks at the local maxima point even if there exists a better maxima point.

This type of problem can be handled by selecting random point of different iterations over times.

We can look over the odds even more clearly from the figure of 'Hill climbing' problem drawn below:

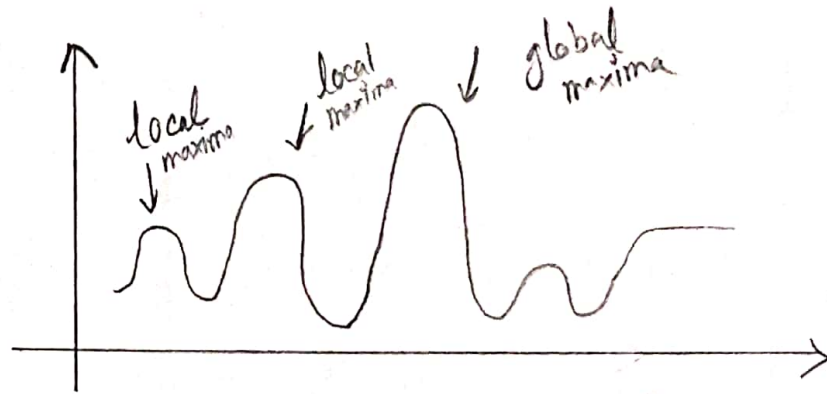


Fig. Hill Climbing Problem

From this figure, we can easily visualize the odds of the environment facing in time of applying optimization technique.

If an intelligent agent wants to climb the maxima point of the hills, then after getting the first maxima point, the agent could stuck at that point thinking the point as maximum point of all hills.

which may hinder the agent to find the actual maxima point of all the hills.

Again,

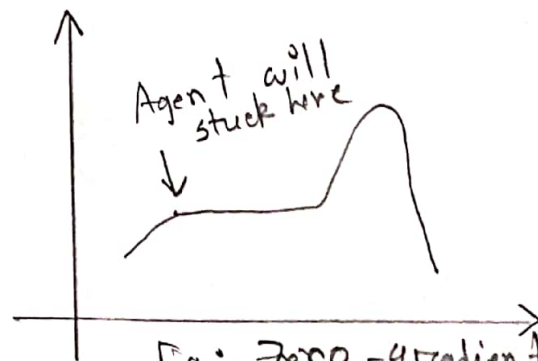
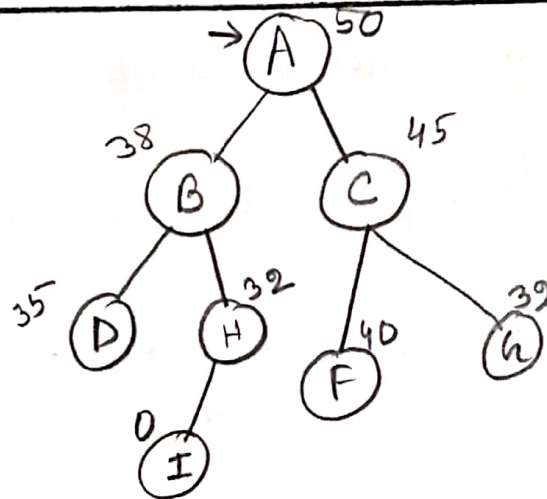


Fig: Zero-gradient problem.

Again, if the agent finds a flat line, it may get stuck at that point as there will be no change of gradient in case of applying optimization technique.

Now, we are describing a situation where a randomly generated successor of a current state having a comparatively lesser value than the other successor will be more promising:



Here, our current state is A, one of the randomly generated successors of A is B which is comparatively lesser than other successor C. Again, for current state B other successor is H which is also lesser than the other successor D.

And finally we get the final successor, I.

After observing the scenario, we can see that selecting random successor of the current state which is comparatively lesser gives us more promising value.

Ans to the Ques No 4 (a)

$$P(V_{12} \cup V_{23}) = P(V_{12}) + P(V_{23})$$

$$= 0.21 + 0.14 + 0.12 + 0.13 + 0.03$$

$$= 0.63$$

$$P(V_{12} | V_{21}) = P(V_{12} \cap V_{21}) / P(V_{21})$$

$$= \frac{0.21}{0.05 + 0.21 + 0.09} = 0.6$$

$$P(V_{22} | V_{12}) = P(V_{22} \cap V_{12}) / P(V_{12})$$

$$= \frac{0.14}{0.21 + 0.14 + 0.12} = 0.298$$

(Ans)

Anstothe Ques NO 4 (b)

Decision theory is the study of an agent's choice to maximize the expected utility.

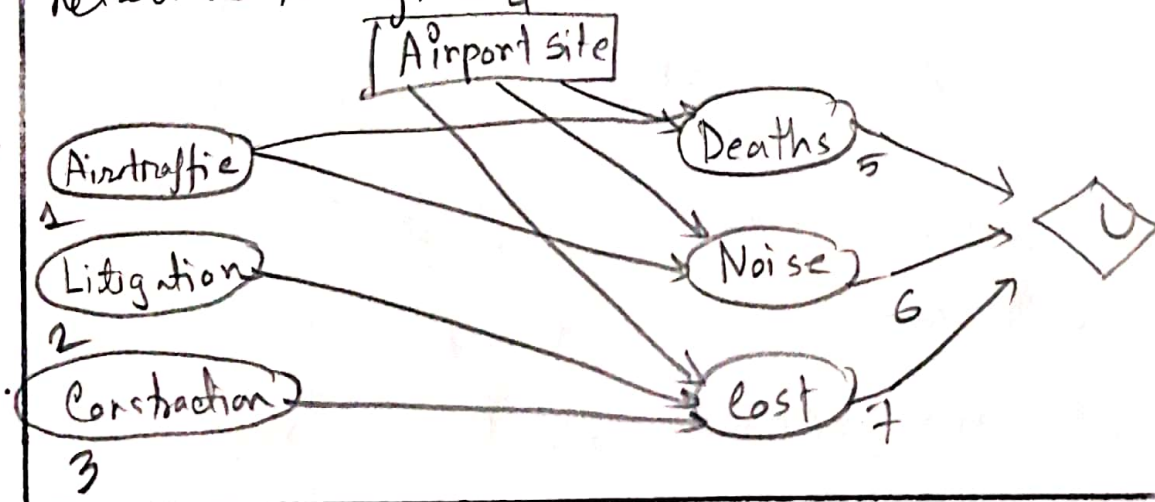
In real world problem an agent is not always informed with total information of its environment. That means a decision theoretic agent has to work with inadequate information. The agent uses probability and utility to outcome an action. An agent combines its belief taken as probability and desire taken as utility to assess the outcome of an action.

A possible way of application of the basic principal of decision theory is

'decision network' -

A decision network is an application of decision theory which is used by a rational agent to make decision using probability and utility. A decision network is an extended Bayesian Network. Decision Networks are also known as influence diagram.

An example of decision network is Airport Siting Problem. A possible network is given below.



In the example above is a network of some decision about death, raise and cost which depend on some other factors like Air traffic, litigation, construction. There are three types of Node in a Decision Network,

① Chance Node: In a chance node, agent is not certain about the situation, we can think of it as a random variable, there are two types of chance decision node.

① current state (1, 2, 3)

② future state (5, 6, 7)

① Decision Node: Here decision are taken by an agent. An agent has alternative at these places. After choosing one it becomes

a chance node. There could be multiple decision node

(14) Utility node: There could be only one utility node in a decision network. It represents the utility function of the agent.